

Anaerobic Co-digestion of Food Waste and Glycerol in Single-Stage System

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List of Abbreviations

AD	A biological process in which microorganisms break down organic matter in the absence of oxygen, producing biogas.
Alk	Buffering capacity of the digester, expressed as $\text{mg CaCO}_3 \text{ L}^{-1}$.
AMPTS II	Automatic Methane Potential Test System
C-glycerol (GL)	crude glycerol
CSTR	A reactor with continuous mixing and feeding, typically used for liquid anaerobic digestion.
FS	fixed solids
FW	food waste
GL	glycerol
HRT	hydraulic retention time
IPB	Instituto Politécnico de Bragança
MONG	matter organic non-glycerol
OLR	organic loading rate
P-GL	pure glycerol
TC	total carbon
TN	total nitrogen
TOC	total organic carbon
TS	total solids
VFA	Intermediate products of anaerobic digestion that can inhibit methanogenesis at high concentrations.
VS	volatile solids
WWTP	wastewater treatment plant

List of Symbols

$\text{Ca}(\text{HCO}_3)_2$	Calcium bicarbonate; buffering agent added to the feed on day 0.
CH_4	Methane.
C/N	Carbon-to-nitrogen ratio of the feed.
CO_2	Carbon dioxide.
$\text{C}_6\text{H}_{12}\text{O}_6$	Glucose; simple sugar added to the feed on day 0 as a readily degradable carbon source.
KHCO_3	Potassium bicarbonate; buffering agent added to the feed.
O_2	Oxygen.
1,3-PDO	1,3-propanediol.
volatile fatty acid (VFA)/alkalinity (Alk)	Ratio of volatile fatty acids to total alkalinity; a stability indicator for anaerobic digesters.
v/v	Volume per Volume

List of Units

M	mega prefix (10^6)
k	kilo prefix (10^3)
m	Milli prefix (10^{-3})
t	metric tonne (10^3 kg)
g	gram
L	litre
m	metre
%	percentage
°C	degrees Celsius
min	minute
h	hour
d	day
yr	year
X	dilution factor (X times)
N mL	normalised millilitre (at standard conditions)
v	Volume
mol dm ⁻³	molarity (mol of solute per litre of solution)
G	relative centrifugal force

Acknowledgements

Abstract

The growing generation of food waste and the surplus of crude glycerol from biodiesel production represent two significant organic waste streams requiring sustainable management solutions. This study evaluated the technical feasibility of a single-stage anaerobic co-digestion system treating food waste and glycerol as co-substrates, with the objective of verifying biogas production while providing a valorisation route for crude glycerol. The investigation was conducted in a 1 L continuous stirred-tank reactor operated under mesophilic conditions (35 °C) with a hydraulic retention time of 30 days. Food waste was collected from a university canteen, and crude glycerol was obtained from biodiesel production residues; anaerobic sludge from a municipal wastewater treatment plant served as inoculum. The reactor was operated in sequential phases: an initial adaptation period, co-digestion of food waste and glycerol, glycerol only feeding, and a post-feeding period. Biogas production was monitored continuously using an automatic methane potential test system, while process stability was assessed through measurements of pH, total alkalinity, volatile fatty acids, and their ratio. Substrate characterisation included total solids, volatile solids, density, total carbon, and total nitrogen analyses. Results demonstrated that co-digestion of food waste and glycerol yielded the highest methane production rates and cumulative volumes, with flow spikes immediately following feeding events indicating rapid substrate conversion. In contrast, glycerol-only feeding resulted in reduced biogas production despite continued methanogenic activity. Total alkalinity remained elevated following bicarbonate supplementation, maintaining pH around 7 throughout most of the experiment; volatile fatty acid concentrations fluctuated between 3000 to 5000 mg CH₃COOH L⁻¹, with VFA/alkalinity ratios of 0.3 to 1.2. These findings confirm the hypothesis that glycerol can be useful for biogas production when co-digested with food waste or alone. The study demonstrates the technical viability of anaerobic co-digestion as a strategy for valorising crude glycerol within a circular bioeconomy framework, though further research is needed to optimise glycerol loading rates and develop robust feeding strategies for industrial application.

Keywords: anaerobic digestion; food waste; crude glycerol; biogas; methane.

Resumo

A crescente geração de resíduos alimentares e o excedente de glicerol bruto proveniente da produção de biodiesel representam duas produções significativas de resíduos orgânicos que requerem soluções sustentáveis. Este estudo avaliou a viabilidade técnica de um sistema de codigestão anaeróbia de uma única etapa, tratando resíduos alimentares e glicerol como co-substratos, com o objetivo de verificar a produção de biogás e, simultaneamente, proporcionar uma via de valorização para o glicerol bruto. A investigação foi conduzida num reator contínuo de tanque agitado de 1 L, operado em condições mesofílicas (35 °C), com um tempo de retenção hidráulica de 30 dias. Os resíduos alimentares foram recolhidos numa cantina universitária e o glicerol bruto foi obtido a partir de resíduos da produção de biodiesel; lamas anaeróbias de uma estação de tratamento de águas residuais municipais serviram como inóculo. O reator foi operado em fases sequenciais: um período inicial de adaptação, codigestão de resíduos alimentares e glicerol, alimentação exclusiva com glicerol e um período pós-alimentação. A produção de biogás foi monitorizada continuamente através de um *automatic methane potential test system*, enquanto a estabilidade do processo foi avaliada por medições de pH, alcalinidade total, ácidos graxos voláteis e respectiva razão. A caracterização dos substratos incluiu análises de sólidos totais, sólidos voláteis, densidade, carbono total e nitrogênio total. Os resultados demonstraram que a codigestão de resíduos alimentares e glicerol originou as taxas de produção de metano e os volumes acumulados mais elevados, com picos imediatamente após as alimentações, indicando uma rápida conversão do substrato. Em contraste, a alimentação exclusiva com glicerol resultou numa produção reduzida de biogás, apesar da atividade metanogénica continuada. A alcalinidade total manteve-se elevada após a suplementação com bicarbonato, mantendo o pH em torno de 7 durante a maior parte da experiência; as concentrações de ácidos graxos voláteis flutuaram entre 3.000 a 5.000 mg CH₃COOH L⁻¹, com razões AGV/alcalinidade de 0,3 a 1,2. Estes resultados confirmam a hipótese de que o glicerol pode ser útil para a produção de biogás quando codigerido com resíduos alimentares ou isoladamente. O estudo demonstra a viabilidade técnica da codigestão anaeróbia como estratégia de valorização do glicerol bruto no âmbito de uma bioeconomia circular, embora seja necessária investigação adicional para otimizar as cargas de glicerol e desenvolver estratégias de alimentação

robustas para aplicação industrial.

Palavras-chave: digestão anaeróbia; resíduos alimentares; glicerol bruto; biogás; metano.

Chapter 1

Introduction

The generation of organic waste has become one of the defining environmental challenges of our time, particularly as urbanization and industrial activity continue to expand. A major contributor is FW, a global stream that in the European Union alone is estimated at 88 Mt yr⁻¹, excluding inedible parts removed from the supply chain (Scherhauser et al., 2018). When considering only post-processing activities – wholesale, retail, food service and households – approximately 66.3 Mt of FW are produced annually in the EU-27+1 (Albizzati et al., 2021). The scale of this loss highlights the urgent need for improved management strategies that move beyond simple disposal.

The environmental burden of FW is substantial. It accounts for roughly 15 % to 16 % of the total climate impact of the entire food supply chain, translating to about 186 Mt CO₂-eq of greenhouse gas emissions, alongside significant contributions to acidification and eutrophication (Scherhauser et al., 2018). The majority of these impacts arise during primary production, making prevention the single most effective strategy; however, when waste cannot be avoided, its valorisation through anaerobic digestion or composting is favoured over landfill disposal (Albizzati et al., 2021; Scherhauser et al., 2018). Despite this, much FW continues to be landfilled or incinerated, forfeiting the opportunity to recover its embedded energy and nutrients.

Parallel to the FW challenge, the rapid expansion of biodiesel production has generated an excess of another organic by-product: C-GL. Formed in the transesterification of triglycerides, approximately 10 kg of C-GL are produced for every 100 kg of biodiesel (Garlapati et al., 2016). Global biodiesel output, driven by renewable energy mandates, has caused the GL market to swell – with C-GL supply projected to reach about 6.33 Mt by 2025, of which nearly 680 kt are expected to be highly impure due to the shift towards waste-based feedstocks (Attarbachhi et al., 2023).

C-GL composition varies widely depending on the oil source and catalyst used, but it typically contains methanol, soaps, free fatty acids, salts and other matter organic non-

glycerol (MONG) compounds (Anand & Saxena, 2012; Attarbachhi et al., 2023). These impurities are known to inhibit microbial growth and fermentation, as demonstrated by the severe reduction in 1,3-PDO production when untreated C-GL from jatropha, soybean or sunflower oils was used (Anand & Saxena, 2012). Purification to technical or pharmaceutical grade is energy-intensive and often uneconomical for small and medium biodiesel plants (Attarbachhi et al., 2023), prompting the search for direct, cost-effective valorisation routes. Because of its high organic content and ready biodegradability, C-GL has attracted attention as an ideal co-substrate for anaerobic digestion, where it can serve as a concentrated carbon source while being simultaneously detoxified by dilution and microbial synergy (Gebreegziabher et al., 2025; Tomczak et al., 2025).

AD is a mature biological process in which consortia of microorganisms convert organic matter into biogas – primarily CH_4 (55 % to 65 %) and CO_2 – under O_2 -free conditions, passing through the sequential stages of hydrolysis, acidogenesis, acetogenesis and methanogenesis (Gebreegziabher et al., 2025). AD offers a unique combination of benefits: it produces renewable, storable energy that can replace fossil fuels; generates a nutrient-rich digestate that can be used as organic fertiliser; and reduces the environmental load of organic wastes by avoiding landfilling and its associated CH_4 emissions (Gebreegziabher et al., 2025; Scherhauser et al., 2018).

Within this framework, anaerobic co-digestion – the simultaneous treatment of two or more substrates – has emerged as a particularly effective strategy. Co-digesting C-GL with nitrogen-rich materials such as animal manure, sewage sludge or FW balances the C/N, dilutes potential inhibitors, provides essential macro- and micronutrients, and fosters synergistic microbial interactions (Gebreegziabher et al., 2025; Tomczak et al., 2025). The result can be a dramatic improvement in biogas and CH_4 yield: studies report increases ranging from 14 % to 226 % when C-GL is added to a variety of base substrates, with CH_4 contents generally exceeding 60 % as long as GL does not exceed about 8 % (v/v) of the feed (Gebreegziabher et al., 2025; Tomczak et al., 2025).

However, excessive GL loads can lead to VFA accumulation, pH drop and methanogen inhibition, underscoring the need for careful optimisation (Gebreegziabher et al., 2025; Tomczak et al., 2025). Overall, the convergence of abundant FW and surplus C-GL supplies, together with the proven capability of AD to convert these streams into renewable CH_4 and fertiliser, represents a tangible route towards a circular, low-carbon bioeconomy.

Chapter 2

Objectives

2.1 General Objectives

The primary objective of this dissertation project is to evaluate the efficiency of a single-stage anaerobic co-digestion system using organic FW and C-GL as substrates. The investigation will be conducted by determining biogas production, with a focus on industrial feasibility.

The central hypothesis is that the controlled addition of C-GL, a by-product of the biodiesel industry, helps biogas production due to its high energy content and rapid biodegradability, provided that adequate operational conditions are maintained to avoid VFA overload.

2.2 Specific Objectives

1. Characterize the substrates (FW and C-GL), evaluating physicochemical parameters relevant to anaerobic digestion.
2. Operate the single-stage anaerobic co-digestion system, assessing process stability.
3. Evaluate the techno-economic feasibility of using C-GL as a co-substrate, considering potential for industrial application.

Chapter 3

Literature Review

The rapid expansion of biodiesel production has generated an oversupply of C-GL, a by-product of the transesterification output, this Fig. 3.1 shows the steps to produce biodiesel and when GL is produced (Garlapati et al., 2016). This waste stream, loaded with impurities such as methanol, soaps, and hydroxides, the Fig. 3.2 shows the processes and what they contaminate the GL with, poses environmental disposal challenges, also with the variation of impurities depending on the source, there's no standard method to use this source of carbon (Anand & Saxena, 2012; Gebreegziabher et al., 2025). AD aligns with circular economy principles by converting organic residues into biogas while recycling nutrients, making C-GL a promising co-substrate (Gebreegziabher et al., 2025).

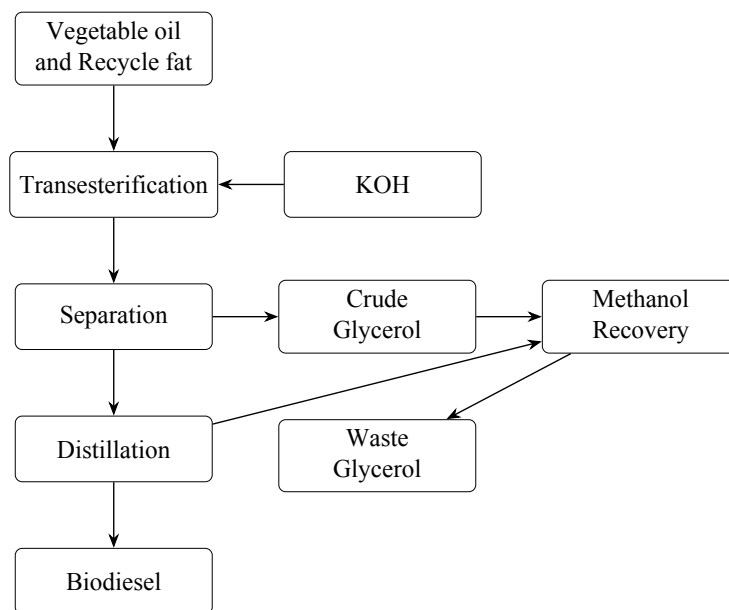


Figure 3.1: crude glycerol from biodiesel production (Adapted from: Garlapati et al., 2016)

Gebreegziabher et al. (2025) reported that co-digestion of C-GL can boost CH_4 production compared to mono-digestion of the base substrate. Tomczak et al. (2025) reviewed

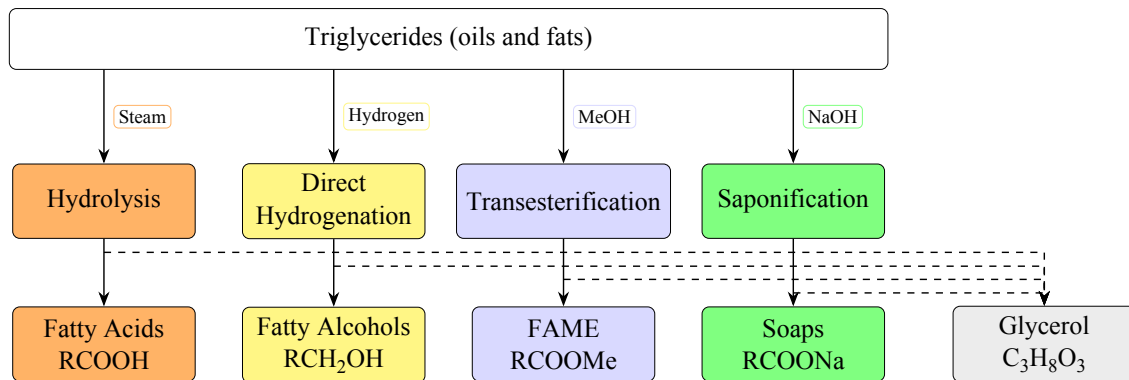


Figure 3.2: Contaminations of the crude glycerol (Adapted from: Attarbach et al., 2023)

one-stage AD experiments and concluded that a CH_4 content above 60 % could be consistently obtained when the C-GL fraction did not exceed 8 % (v/v) of the feed, the flow chart by Li et al. (2011) on the Fig. 3.3 it shows that using AD there could be up to 70 % CH_4 production. However, the same review emphasized that increasing the GL loading beyond a system-specific threshold may provoke inhibitory effects, likely due to the accumulation of VFAs and the impurities present in C-GL (Anand & Saxena, 2012; Tomczak et al., 2025). The heterogeneous composition of C-GL, affected by oil feedstock, catalyst, and biodiesel processing, demands careful feedstock characterization and, if necessary, simple pretreatment to avoid process upsets (Attarbach et al., 2023).

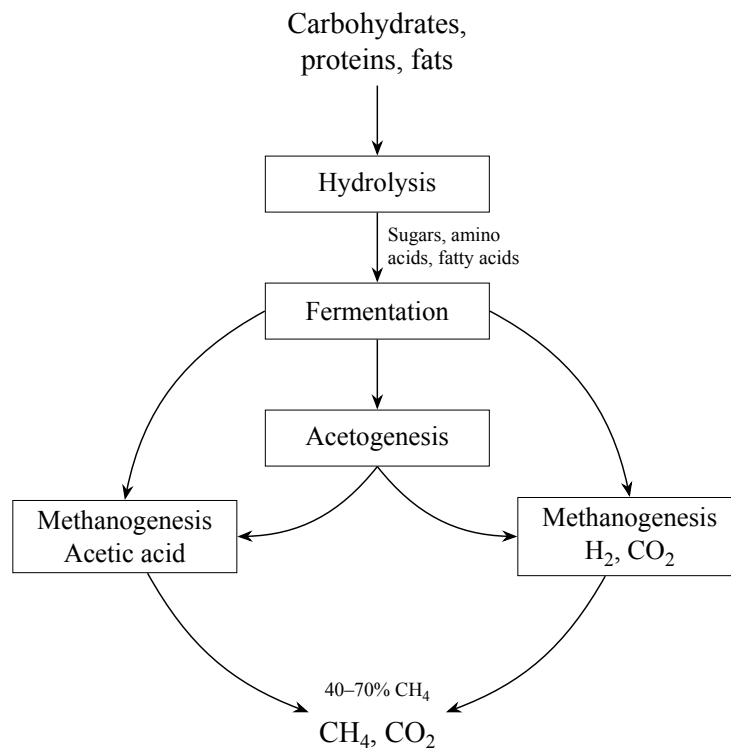


Figure 3.3: CH_4 production from AD (Adapted from: Li et al., 2011)

Liquid AD, typically conducted in a continuous stirred-tank reactor (CSTR), operates at total solid concentrations of 0.5 % to 15 % and is generally used for treating sewage sludge, manure, and FW (Li et al., 2011). The CSTR's continuous mixing provides uniform distribution of substrates, microorganisms, and heat, which is essential when feeding a fermentable substrate. Temperature is a critical operational parameter; the activity of methanogenic consortia is strongly temperature dependent, and both mesophilic and thermophilic regimes have been applied to AD (Choorit & Wisarnwan, 2007). Tomczak et al. (2025) noted that the majority of GL co-digestion studies have been performed under mesophilic conditions, while thermophilic operation, which could offer higher conversion rates, still requires more systematic investigation. The CSTR platform is well suited to explore such temperature effects because its design allows easy control of heating and hydraulic retention time.

The selection of a robust microbial inoculum is decisive for rapid start-up and stable performance of a CSTR fed with GL. Anaerobic sludge from municipal wastewater treatment plants (WWTPs) is the most common source of inoculum because it harbours a diverse consortium of hydrolytic, acidogenic, and methanogenic microorganisms, the flow chart from Gebreegziabher et al. (2025) on the Fig. 3.4 shows each step of the AD until the CH₄ production. Posmanik et al. (2020) demonstrated that granular sludge, obtained from up-flow anaerobic sludge blanket reactors treating wastewater, outperforms suspended biomass as an inoculum. Although their work focused on batch tests, the findings strongly suggest that using granular sludge from a WWTP as seed for a CSTR could accelerate acclimation, improve process stability, and reduce the risk of acidification during load increases.

When GL starts being degraded it makes the AD process susceptible to VFA accumulation and pH drop if the organic loading rate is not properly controlled. Therefore, close monitoring of process indicators is essential. Ribas et al. (2007) evaluated several methods for determining these parameters specifically in anaerobic reactors, highlighting the importance of accurate VFA and Alk measurements for timely detection of process imbalance. Using the method from DiLallo & Albertson (Ribas et al., 2007) provide a validated method for measuring VFAs and Alk. Maintaining a favourable VFA/Alk is particularly critical when co-digesting GL, as excess VFAs can inhibit methanogenesis (Tomczak et al., 2025).

The available literature confirms that C-GL is a high-energy substrate that can significantly boost CH₄ production in liquid AD systems. A CSTR provides the mixing and temperature control required to harness this potential. Using granular sludge from a WWTP as the inoculum can enhance start-up speed and methanogenic activity, as demonstrated by comparative studies. To unlock the full benefits while avoiding inhibition, the GL loading

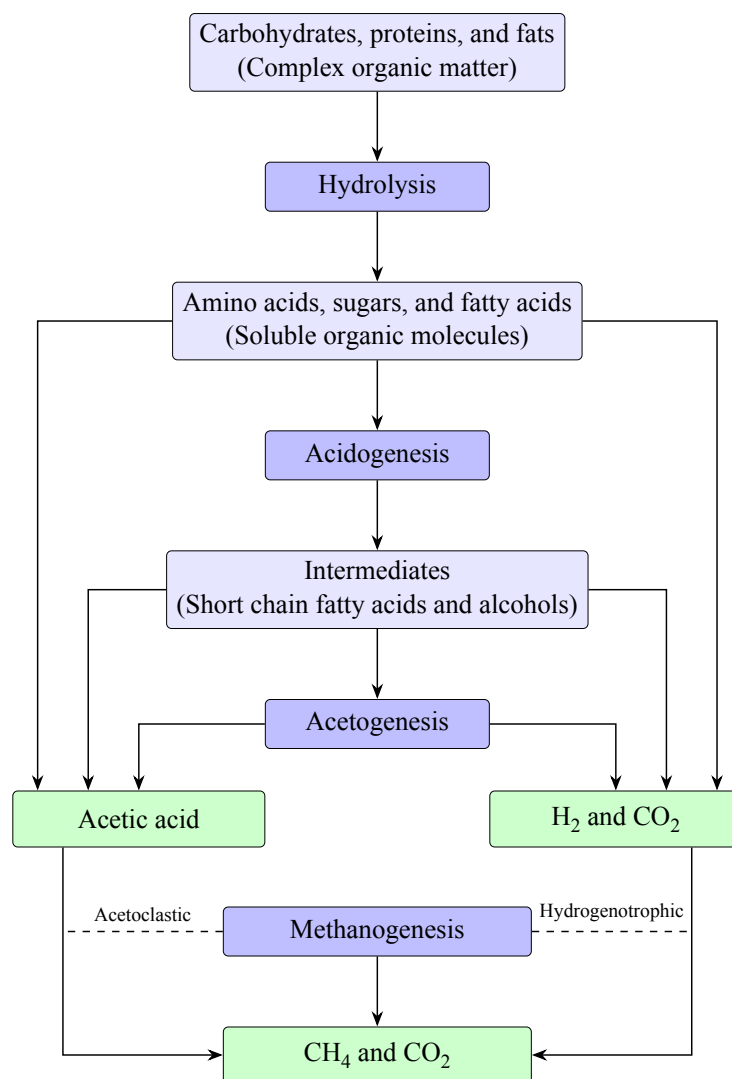


Figure 3.4: Steps to produce CH_4 from AD (Adapted from: Gebreegziabher et al., 2025)

must be kept below documented thresholds, and the reactor state should be continuously tracked through standard measurements of VFAs and Alk. Further research is needed to optimise thermophilic CSTR operation and to establish long-term performance data for granular-sludge-inoculated reactors treating C-GL.

Chapter 4

Materials and Methods

4.1 Collection and Preparation of the Food Waste

FW was collected from the canteen at the Instituto Politécnico de Bragança (IPB), prioritizing residues with high proportions of rice, pasta, and bread to maximize the sugar content and minimize the nitrogen content. The waste was diluted with approximately 2.5 L of distilled water and homogenised using a blender to obtain a liquid paste. It was then stored in a refrigerator at 4 °C to prevent spoilage and suppress microbial and fungal activity (Degueurce et al., 2020).

4.2 Crude Glycerol Collection

A 6 L bottle of C-GL, obtained from previous biodiesel production experiments at IPB, was provided for this study. C-GL from biodiesel typically contains impurities such as catalyst residues (Attarbach et al., 2023), which may influence its behaviour in anaerobic digestion.

4.3 Inoculum Collection

The anaerobic inoculum consisted of sludge collected from Águas do Norte wastewater treatment plant in Bragança. This ensured the availability of a microbial population for the anaerobic digestion (Posmanik et al., 2020).

4.4 Physical and Chemical Characterisation

The total solids (TS), volatile solids (VS), fixed solids (FS), and density were determined for the FW, C-GL, and inoculum. Total organic carbon (TOC) and total nitrogen (TN) were determined for the FW and C-GL.

4.4.1 TS and VS Determination

TS and VS were measured using a sequential gravimetric method based on Standard Methods 2540 B and 2540 E (Baird et al., 2017). The method involves controlled drying and calcination steps to quantify TS and VS fractions.

Initially, porcelain crucibles were placed in a muffle furnace at 550 °C for 1 h. For cooling, they were transferred to an oven at 105 °C for 15 min and then moved to a desiccator until they reached room temperature; their empty masses were recorded. For each material, a sample was placed into three crucibles and the wet mass was measured. The crucibles were then dried in an oven at 105 °C for 2 h (C-GL required 6 h to 8 h to reach constant mass). After drying, the crucibles were cooled in a desiccator to room temperature and weighed again to determine TS.

The same crucibles were subsequently placed in a muffle furnace at 550 °C for 50 min. Afterwards they were transferred to an oven at 105 °C for 15 min and then moved to a desiccator until they reached room temperature; the final masses were recorded to calculate VS.

TS, VS and FS were calculated as follows:

$$TS = \frac{M_{105} - M_{crucible}}{V_{sample}} \quad [g \ L^{-1}] \quad (4.1)$$

$$VS = \frac{M_{105} - M_{550}}{V_{sample}} \quad [g \ L^{-1}] \quad (4.2)$$

$$FS = \frac{M_{550} - M_{crucible}}{V_{sample}} \quad [g \ L^{-1}] \quad (4.3)$$

where M_{wet} is the mass of crucible plus wet sample, M_{105} is the mass after drying at 105 °C, M_{550} is the mass after ignition at 550 °C, $M_{crucible}$ is the mass of the empty crucible and V_{sample} is the volume of the sample. The volume is in L and the weight is in g. The results are on the Table 4.1.

Table 4.1: Total solids, Volatile solids and Fixed solids values

Sample		M_{crucible} (g)	M_{wet} (g)	M_{sample} (g)	V_{sample} (L)	M_{105} (g)	TS (g L ⁻¹)	Average TS (g L ⁻¹)	M_{550} (g)	VS (g L ⁻¹)	Average VS (g L ⁻¹)	FS (g L ⁻¹)	Average FS (g L ⁻¹)
Food Waste	1.000	7.873e1	1.060e2	2.726e1	2.550e-2	9.100e1	4.810e2		7.920e1	4.625e2		1.848e1	
	2.000	8.432e1	1.133e2	2.898e1	2.710e-2	9.887e1	5.364e2	5.364e2	8.474e1	5.208e2	5.208e2	1.559e1	1.848e1
	3.000	7.759e1	1.088e2	3.125e1	2.920e-2	9.344e1	5.420e2		7.818e1	5.216e2		2.038e1	
Crude Glycerol	1.000	8.422e1	8.864e1	4.424	5.000e-3	8.850e1	8.634e2		8.421e1	8.644e2		-9.891e-1	
	2.000	7.348e1	7.799e1	4.509	5.000e-3	7.786e1	8.677e2	8.677e2	7.347e1	8.687e2	8.687e2	-9.308e-1	-9.891e-1
	3.000	8.620e1	9.061e1	4.410	4.900e-3	9.050e1	8.713e2		8.620e1	8.724e2		-1.094	
Inoculum	1.000	7.873e1	1.060e2	2.726e1	2.550e-2	9.100e1	4.810e2		7.920e1	4.625e2		1.848e1	
	2.000	8.432e1	1.133e2	2.898e1	2.710e-2	9.887e1	5.364e2	5.364e2	8.474e1	5.208e2	5.208e2	1.559e1	1.848e1
	3.000	7.759e1	1.088e2	3.125e1	2.920e-2	9.344e1	5.420e2		7.818e1	5.216e2		2.038e1	

4.4.2 Density Determination

Density was determined by a gravimetric–volumetric method using graduated cylinders. Dry cylinders were weighed empty, then filled with distilled water at 20 °C and weighed again to verify the reading accuracy (density of water at $\rho_{20^{\circ}\text{C}} = 998.21 \text{ g L}^{-1}$). After drying, each cylinder was filled with a known volume of sample and weighed. The density of each material was calculated using

$$\rho = \frac{M_{\text{total}} - M_{\text{cylinder}}}{V_{\text{sample}}} \quad [\text{g L}^{-1}] \quad (4.4)$$

where M_{total} is the mass of cylinder plus sample, M_{cylinder} is the empty cylinder mass, and V_{sample} is the measured volume, the masses were measured in g and the volume in L. The results are on the Table 4.2.

Table 4.2: Density values

Sample		M_{cylinder} (g)	M_{total} (g)	V_{sample} (L)	Density (g L^{-1})	Average Density (g L^{-1})
Food	1.000	1.360e2	2.420e2	9.920e−2	1.069e3	1.068e3
Waste	2.000	1.160e2	2.230e2	1.002e−1	1.068e3	
Crude	1.000	1.142e2	2.035e2	9.990e−2	8.932e2	8.931e2
Glycerol	2.000	1.157e2	2.044e2	9.930e−2	8.929e2	
Inoculum	1.000	1.360e2	2.420e2	9.920e−2	1.069e3	1.068e3
	2.000	1.160e2	2.230e2	1.002e−1	1.068e3	

4.4.3 TOC and TN Determination

TOC, total carbon (TC) and TN were measured using a Shimadzu TOC-5000A analyser. Prior to analysis, the FW was diluted 400 X and the C-GL was diluted 800 X to bring the concentrations within the linear range of the instrument. The results are on the Table 4.3.

Table 4.3: Total carbon and Total nitrogen values

Sample		TOC (mg L^{-1})	TC (mg L^{-1})	Average TC (mg L^{-1})	TN (mg L^{-1})	Average TN (mg L^{-1})
Food	1.000	5.039	2.018e3	1.779e3	2.756e2	2.818e2
Waste	2.000	3.927	1.540e3		2.879e2	
Crude	1.000	8.808	7.050e3	5.523e3	4.700e2	4.886e2
Glycerol	2.000	4.992	3.997e3		5.072e2	

4.5 Experiment Configuration

The reactor had a useful volume of 1 L and was operated as a CSTR under anaerobic conditions. Mixing was provided by a magnetic stir bar, and the overflow exit tube was kept closed with a valve to maintain an anaerobic environment. As you can see in this Fig. 4.1:

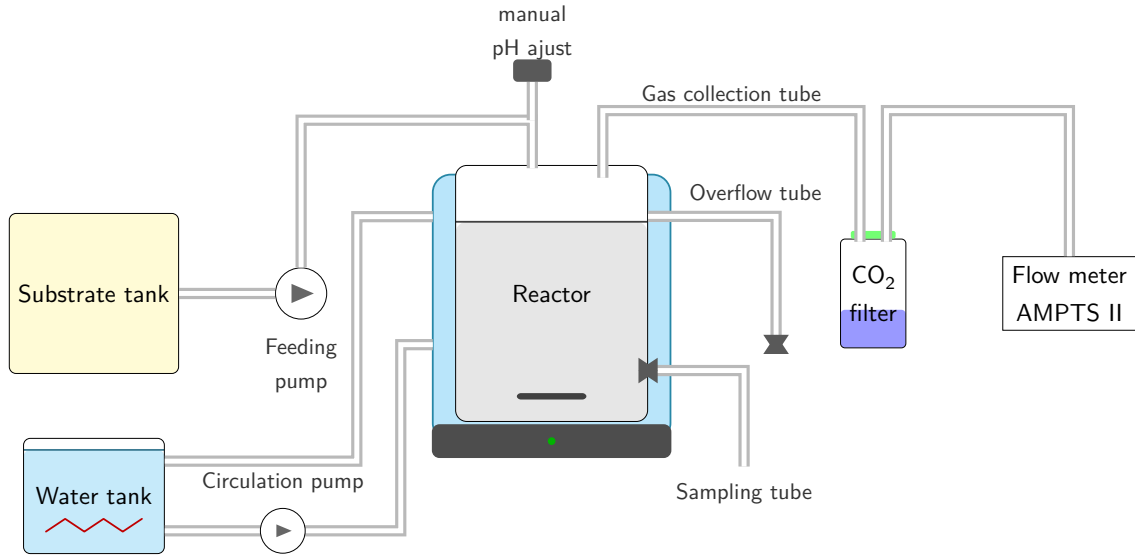


Figure 4.1: Model of the system used

The reactor was kept at 35 °C, close to the optimal temperature for mesophilic microorganisms (Choorit & Wisarnwan, 2007). The organic loading rate (OLR) was set with a max of 4 kg VS m⁻³ d⁻¹, with a hydraulic retention time (HRT) of 30 days, conditions for stable co-digestion of FW and GL (Tomczak et al., 2025). The pH was maintained around 7 by incorporating KHCO₃ into the feed solution at a concentration of 10 g L⁻¹, and adding KHCO₃ to the reactor whenever the pH dropped lower than 6.

The OLR and HRT:

$$\text{HRT} = \frac{V}{Q} \quad [\text{d}] \quad (4.5)$$

$$\text{OLR} = \frac{S_{\text{in}} \times Q}{V} \quad [\text{kg VS m}^{-3} \text{ d}^{-1}] \quad (4.6)$$

where S_{in} is the influent VS concentration (kg m⁻³), Q is the volumetric flow rate (m³ d⁻¹), and V is the reactor working volume (m³).

4.5.1 Feeding Solution

The feed was designed to achieve a C/N ratio of 25:1, which is optimal for AD (Li et al., 2011). Characterisation of the FW and C-GL showed that both substrates contained nitrogen, making it difficult to balance the C/N ratio using both of them together. Therefore, pure glycerol (P-GL) was used as an additive to allow precise control of the C/N. Based on the chosen HRT and the daily volumetric flow rate, the reactor was fed once per day to ensure a flow rate high enough to transport the solid FW suspension through the feeding tube.

The feed was made with a restriction of needing to have less than 15 % of TS%. The amount of OLR and if KHCO_3 was added, was decided by the pH and how was the CH_4 production of the reactor. If it was too acidic, it was used a lower OLR with KHCO_3 , if it was at the right amount it was keep the same amount, and if it was too basic it was added a higher OLR without KHCO_3 . The feed storage always had some remaining feed, and the new one was added on top, diluting or concentrating the feed going to the reactor.

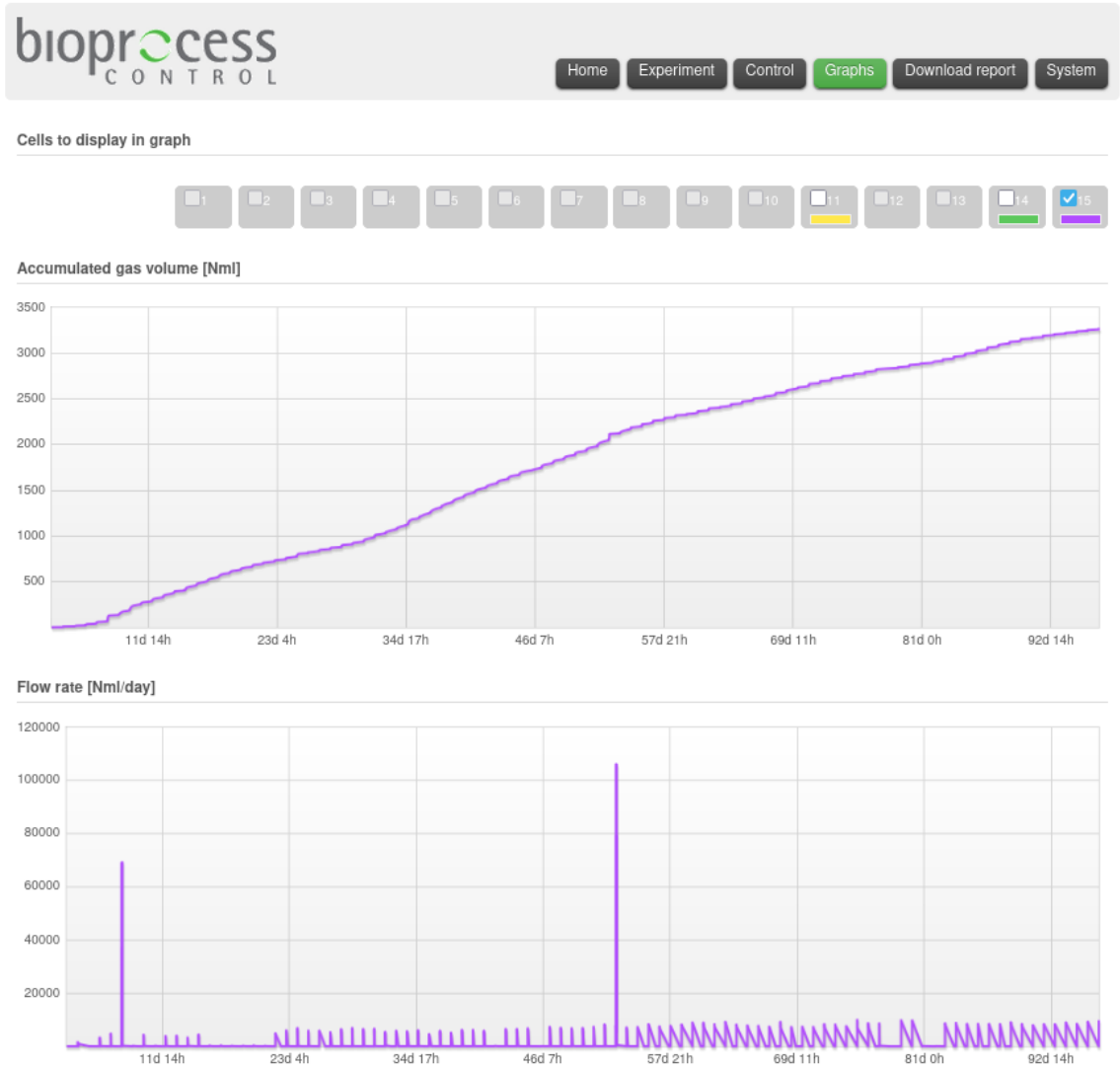
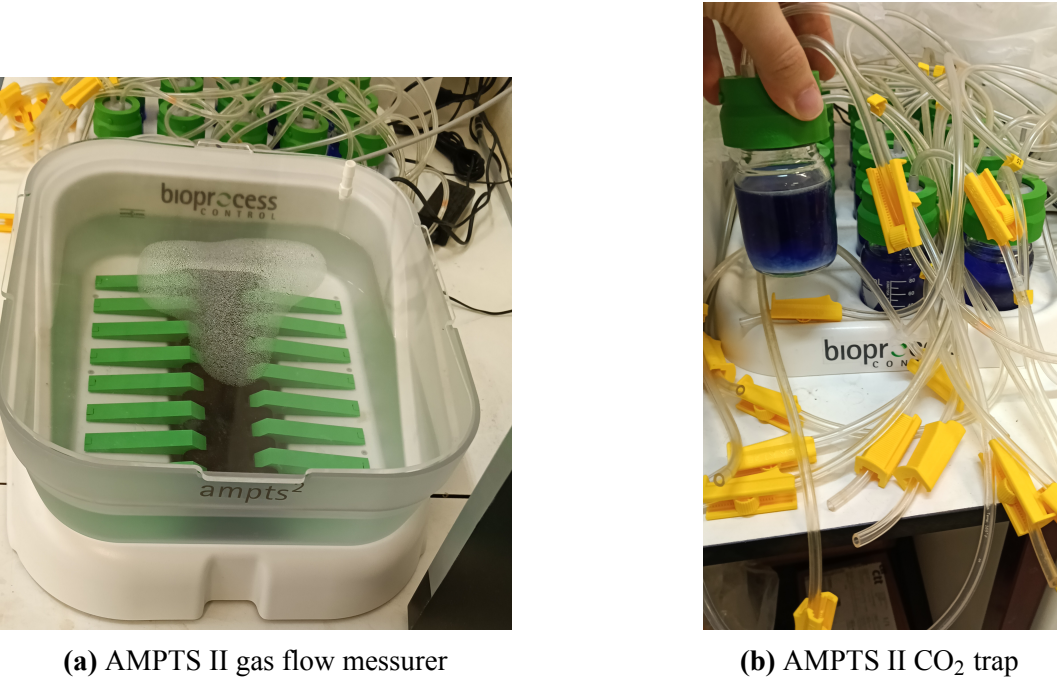
The feed was also supplemented with nutrients to complement the feed and create a better feed for the microorganisms. Table 4.4 shows the nutrients used and there ratios.

Table 4.4: Nutrients used and there ratios, for a 25:1 C/N. (Adapted from: Alqudah, 2025)

Nutrients	Mass (mg (350 mL) ⁻¹)	Mass (mg L ⁻¹)
$\text{MgSO}_4 \cdot 7 \text{H}_2\text{O}$	1.890e1	5.400e1
$\text{FeSO}_4 \cdot 7 \text{H}_2\text{O}$	5.228e2	1.494e3
$\text{CoCl}_2 \cdot 6 \text{H}_2\text{O}$	4.200e-1	1.200
ZnCl_2	1.050e1	3.000e1
$\text{NiCl}_2 \cdot 6 \text{H}_2\text{O}$	1.050e-2	3.000e-2
EDTA	2.100e-1	6.000e-1
$\text{CuSO}_4 \cdot 5 \text{H}_2\text{O}$	9.230	2.637e1

4.5.2 Biogas Quantification

The AMPTS II system (Fig. 4.2), was provided to measure the gas flow that was coming out of the reactor, with a CO_2 fixing unit using a solution of $3 \text{ mol dm}^{-3} \text{ NaOH}$.



(c) AMPTS II system

Figure 4.2: AMPTS II system used

4.5.3 Analysis Methods

It was used the method of DiLallo & Albertson (Ribas et al., 2007), to measure the Alk and VFA. The sample collected was first centrifuged, at 2604 G for 30 min, to filter the particulated before doing the measurements.

$$TA = \frac{N_{acid} * V_{acid} * 50000}{V_{sample}} \quad [\text{mg CaCO}_3 \text{ L}^{-1}] \quad (4.7)$$

$$VFA = \frac{N_{base} * V_{base} * 60000}{V_{sample}} \quad [\text{mg CH}_3\text{COOH L}^{-1}] \quad (4.8)$$

The volumes were measured in mL, the sample volume was always 50 mL, the normality (N) of the acid and base, at first was 0.1, but then it was changed to 1. The Equation (4.7) is expressed in mg CaCO₃ L⁻¹, and the Equation (4.8) is expressed in mg CH₃COOH L⁻¹.

4.5.4 Substrate used as feed

Table 4.5 shows all substrates used to feed the reactor and when they were made.

Table 4.5: Substrates used during the process

Day	Substrate 1	Substrate 2	Substrate 3	Extra Addition	OLR & TS%	Volume Made
0 ^a	FW, 156 g	C ₆ H ₁₂ O ₆ , 1.9 g		Ca(HCO ₃) ₂ , 5 g	1.5 & 15	500 mL
4	C-GL, 23.2 mL	P-GL ^b , 31.5 mL		KHCO ₃ , 1.9 g	0.8 & 10	250 mL
11	FW, 12.54 g	C-GL, 9.02 mL	P-GL, 2.6 mL	KHCO ₃ , 1.57 g	0.13 to 0.4 & 5 to 10 ^c	250 mL
18	FW, 10.88 g	C-GL, 2.89 mL	P-GL, 1.44 mL		0.0256 & 2	250 mL
25	FW, 31.12 g	C-GL, 44 mL	P-GL, 29 mL		1.4 & 10	250 mL
32	FW, 31.01 g	C-GL, 44 mL	P-GL, 28.9 mL		1.4 & 10	250 mL
39	FW, 46.52 g	C-GL, 66 mL	P-GL, 43 mL	KHCO ₃ , 2.6 g	3.1 & 15	250 mL
46	FW, 50.07 g	C-GL, 66 mL	P-GL, 43 mL	KHCO ₃ , 2.67 g	3.16 & 15	250 mL
53 ^d	C-GL, 55 mL	P-GL, 11.5 mL		KHCO ₃ , 5.38 g	0.42 & 10	500 mL

Table 4.5 continued

Day	Substrate 1	Substrate 2	Substrate 3	Extra Addition	OLR & TS%	Volume Made
60	C-GL, 55 mL	P-GL, 12 mL		KHCO ₃ , 5.18 g	0.42 & 10	500 mL ^e
67	C-GL, 41 mL	P-GL, 9 mL		KHCO ₃ , 2.56 g	0.95 & 15	250 mL
74	C-GL, 60 mL	P-GL, 12.5 mL		KHCO ₃ , 2.57 g	1.37 & 15	250 mL

^a It only started feeding on day 2.

^b The solution of P-GL was diluted 100 X.

^c The FW wasn't well mixed this day.

^d Change of substrate.

^e There was a leak on the feeding tube.

Chapter 5

Results

5.1 Reactor Condition and Visual Observations

The reactor was inoculated with anaerobic sludge and initially contained a murky, light brown suspension (Fig. 5.1a). Within three days of operation, the colour changed to black (Fig. 5.1b), which is characteristic of active anaerobic digestion and is typically caused by the precipitation of metal sulphides, particularly iron sulphide, under reducing conditions. This visual change indicated the establishment of a suitable anaerobic environment and the onset of methanogenic activity.



(a) Murky suspension



(b) Black suspension

Figure 5.1: Visual changes in the reactor.

At the end of the experimental period, after approximately two weeks of unattended operation, four discrete masses resembling fungal colonies were observed on the internal surfaces of the reactor during cleaning (Figs. 5.2a and 5.2b). The origin of these growths is uncertain, but they may have been introduced via the FW feed or during periods when

the feeding tube was opened for repairs (e.g. leaks) and when changing the feed that the tube had to be cleaned. The presence of such organisms could have competed with the anaerobic microbial community for substrate, potentially affecting biogas production in the later stages of the experiment.



(a) Masses inside the dry reactor



(b) Masses floating in the reactor

Figure 5.2: Visual for the masses inside the reactor.

5.2 Biogas Production

Biogas production was monitored continuously using AMPTS II, which recorded both the cumulative volume of CH_4 produced and the instantaneous flow rate. The entire experimental period is summarised in Figs. 5.3 and 5.4. To facilitate interpretation, the data have been divided into four distinct phases based on the feeding regime: (i) an adaptation phase 5.2.2, (ii) co-digestion of food waste and glycerol 5.2.3, (iii) digestion of crude glycerol 5.2.4, and (iv) a post-feeding period 5.2.5.

5.2.1 Overall Production

The cumulative CH_4 production over the whole experiment (Fig. 5.3) shows a continuous increase, indicating that the reactor maintained methanogenic activity throughout the monitored period. The total flow profile (Fig. 5.4) exhibits distinct peaks corresponding to daily production of CH_4 , with variations in amplitude reflecting the different substrate compositions and organic loading rates applied during each phase. And a outlier peak at day 53, that was when the feed was changed from a mixture of FW and C-GL, to only C-GL, and due to some complications, created an anomaly in the graph.

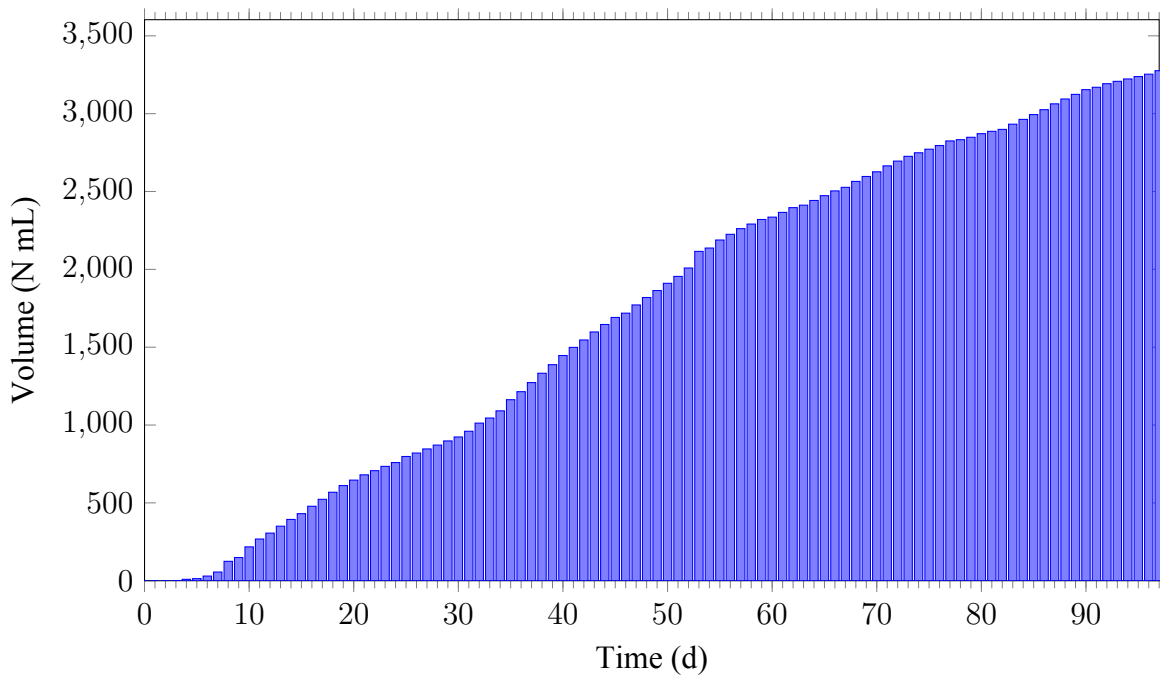


Figure 5.3: Cumulative volume of CH₄ N mL

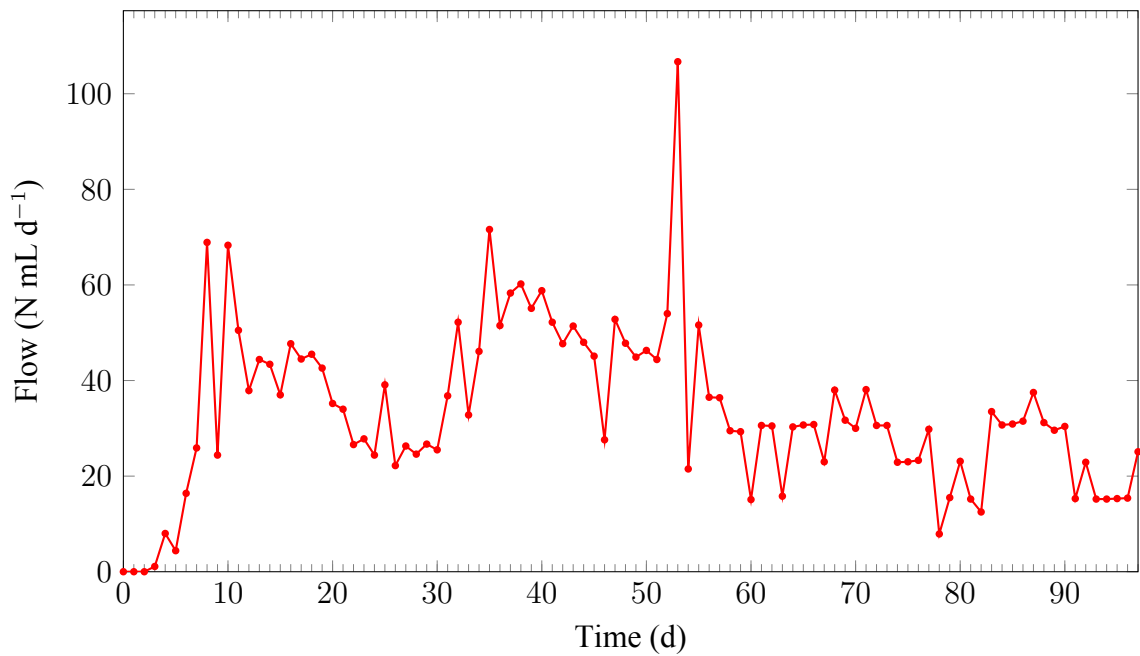


Figure 5.4: Volume flow of CH₄ N mL d⁻¹

5.2.2 Adaptation Phase

During the initial adaptation phase, biogas production was low (Figs. 5.5 and 5.6). This lag period is normal for anaerobic reactors as the microbial community adjusts to the new substrate and operating conditions. The small but measurable CH_4 flow confirmed that the viability of the inoculum, and the gradual increase in flow suggested the onset of exponential growth of the microorganisms.

The spikes seen on the days 8 and 10 (Fig. 5.6) probably happened by the first addition of C-GL to the feed.

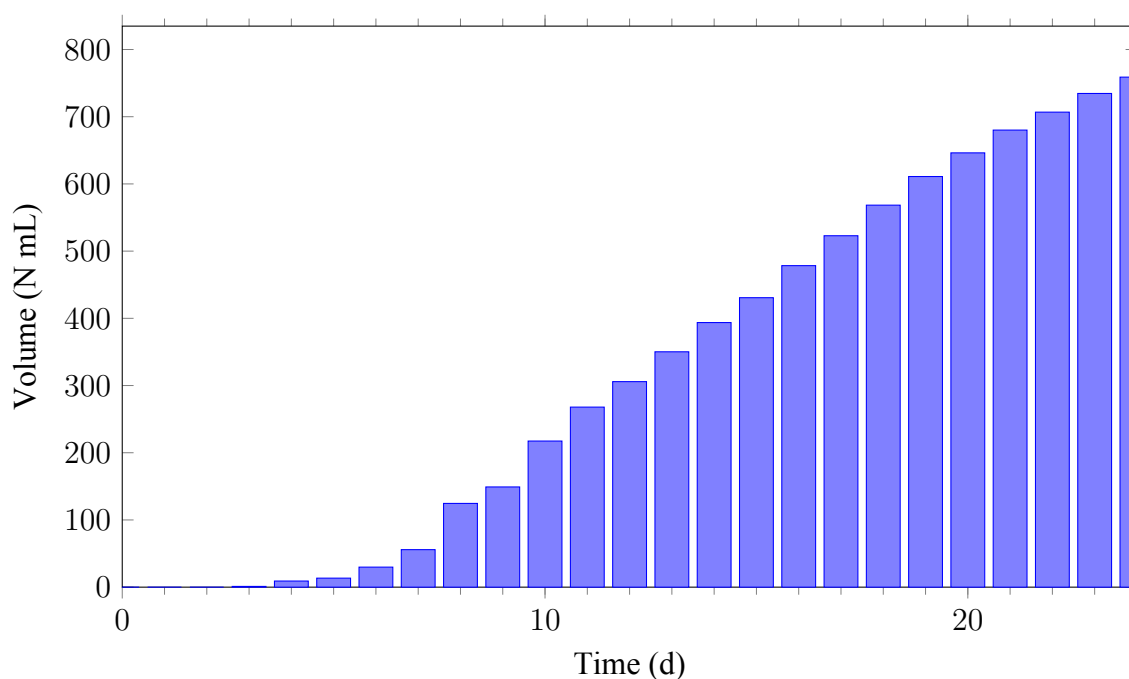


Figure 5.5: Cumulative volume of CH_4 N mL (Adaptation Phase)

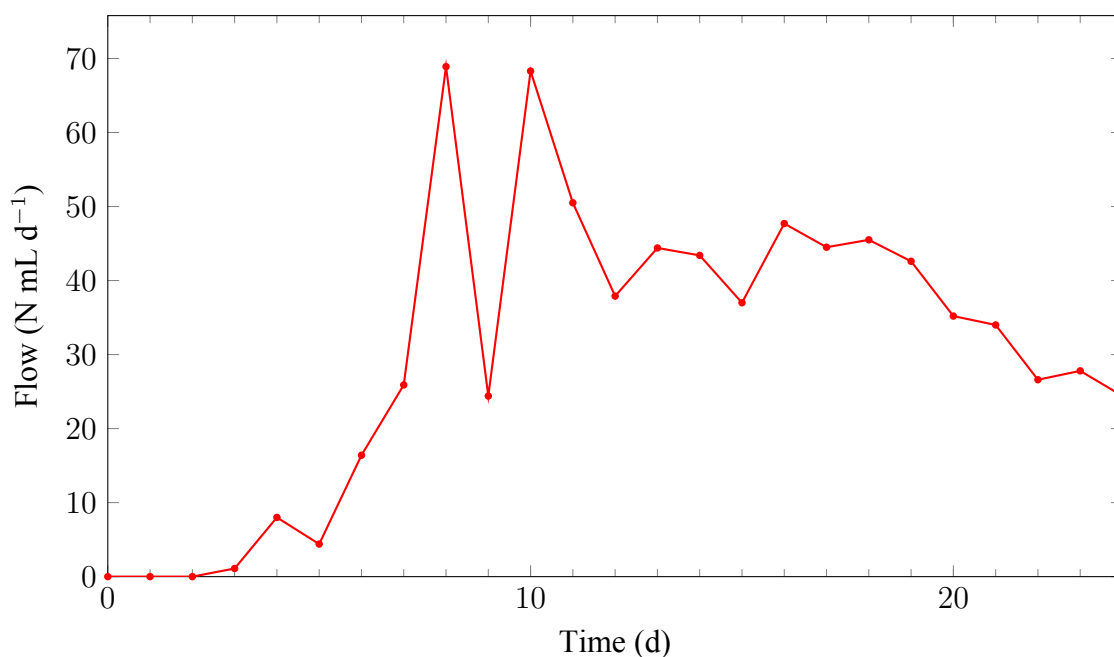


Figure 5.6: Volume flow of CH₄ N mL d⁻¹ (Adaptation Phase)

5.2.3 Co-digestion of Food Waste and Crude Glycerol

The reactor was fed with a mixture of FW and C-GL, a marked increase in biogas production was observed (Figs. 5.7 and 5.8). The cumulative CH₄ volume rose steeply, and the flow rate showed pronounced spikes immediately after each feeding event (Fig. 5.9). These spikes are typical of readily degradable substrates, which are quickly hydrolysed and converted to VFA and subsequently to CH₄. The average flow rate during this phase was higher than in any other phase, indicating that the co-digestion of FW and C-GL provided a balanced nutrient supply and avoided the inhibition often associated with high GL concentrations alone.

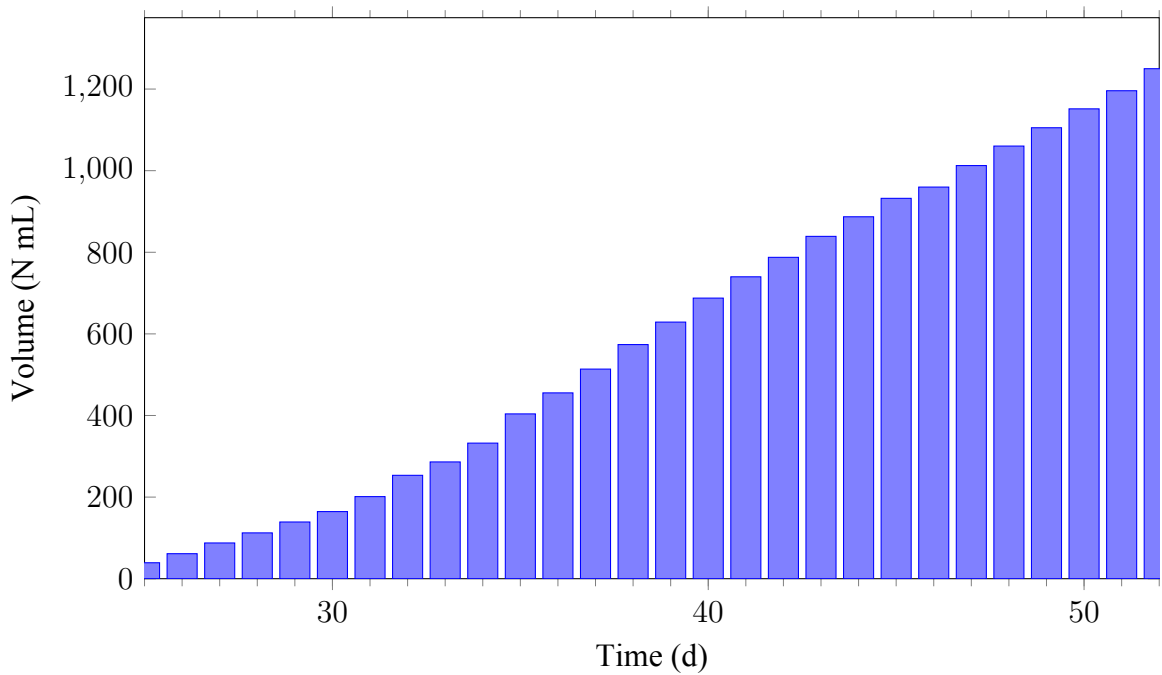


Figure 5.7: Cumulative volume of CH₄ N mL (co-digestion of FW and C-GL)

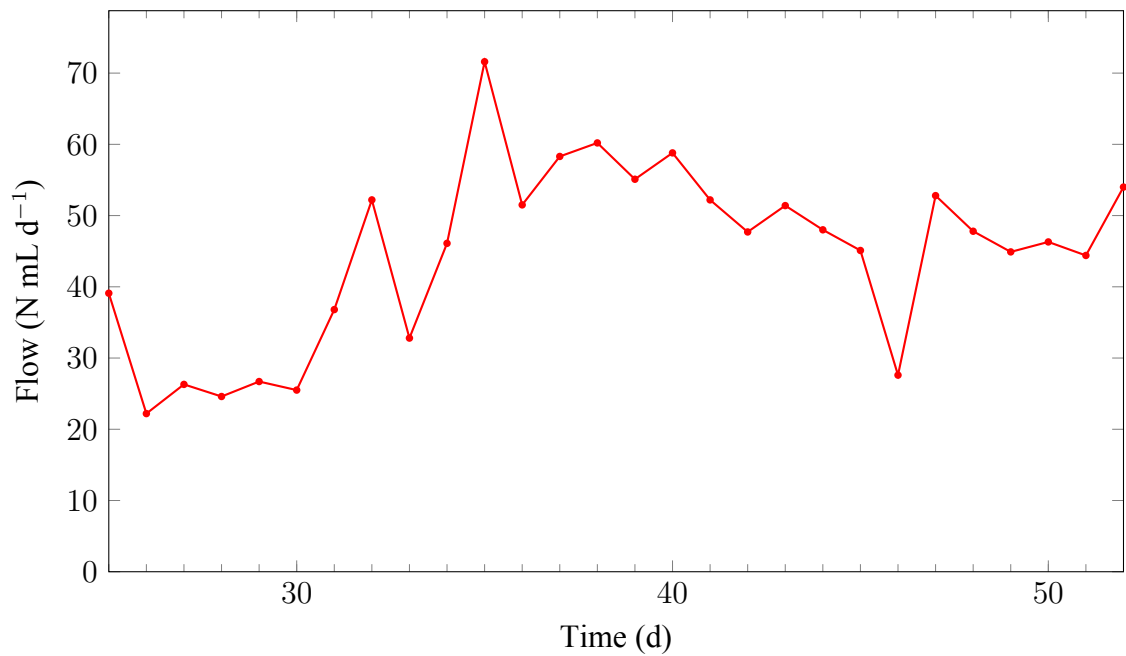


Figure 5.8: Volume flow of CH₄ N mL d⁻¹ (co-digestion of FW and C-GL)

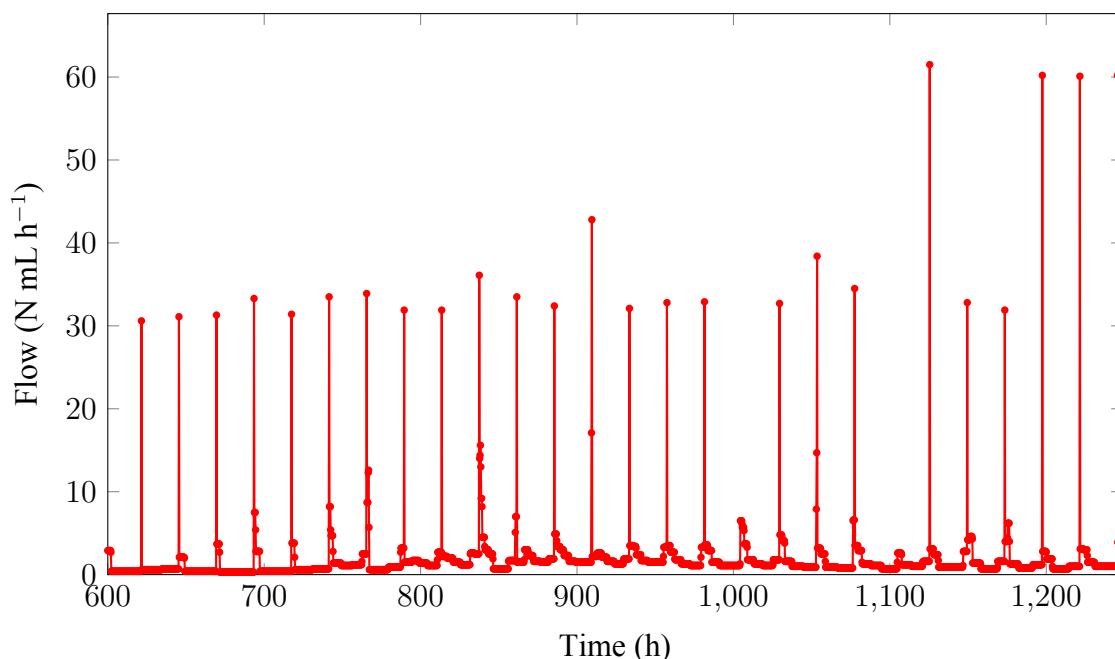


Figure 5.9: Volume flow of CH_4 N mL h^{-1} (co-digestion of FW and C-GL)

5.2.4 Digestion of Crude Glycerol

When the feed was changed to C-GL alone, biogas production decreased (Figs. 5.10 and 5.11). Although spikes in the flow were still observed after feeding (Fig. 5.12), their magnitude was bigger than during the co-digestion phase, and the cumulative CH_4 volume plateaued more quickly. Several factors may explain this lower performance:

- The GL could have been transformed too fast or too slow, to maintaining a concentration of intermediate products.
- The retention rate of $\frac{1}{30} \text{ d}^{-1}$ may have been insufficient to wash out inhibitory by-products or to maintain a stable population of glycerol-fermenting bacteria.
- The appearance of fungal masses towards the end of this phase could have further stressed the anaerobic community by competing for substrate or introducing toxic metabolites. The fungal masses could have helped digest more complex sugars, so without a more thorough investigation of what it was, it can't be said it didn't help the AD system.

For those factors, can't be said that this phase is worse than the co-digestion phase. If the experiment was done in reverse, it's likely that we would have seen a reverse on the results, were the phase with only C-GL would have a bigger production than the co-digestion of FW and C-GL.

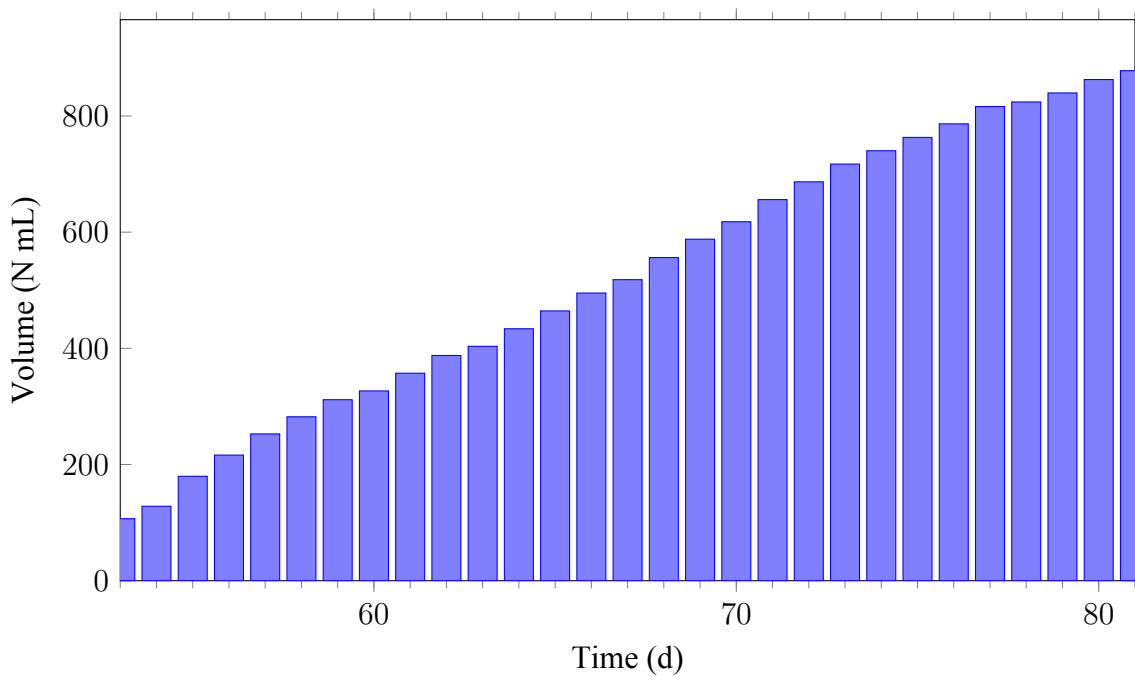


Figure 5.10: Cumulative volume of CH_4 N mL (Anaerobic digestion of C-GL)

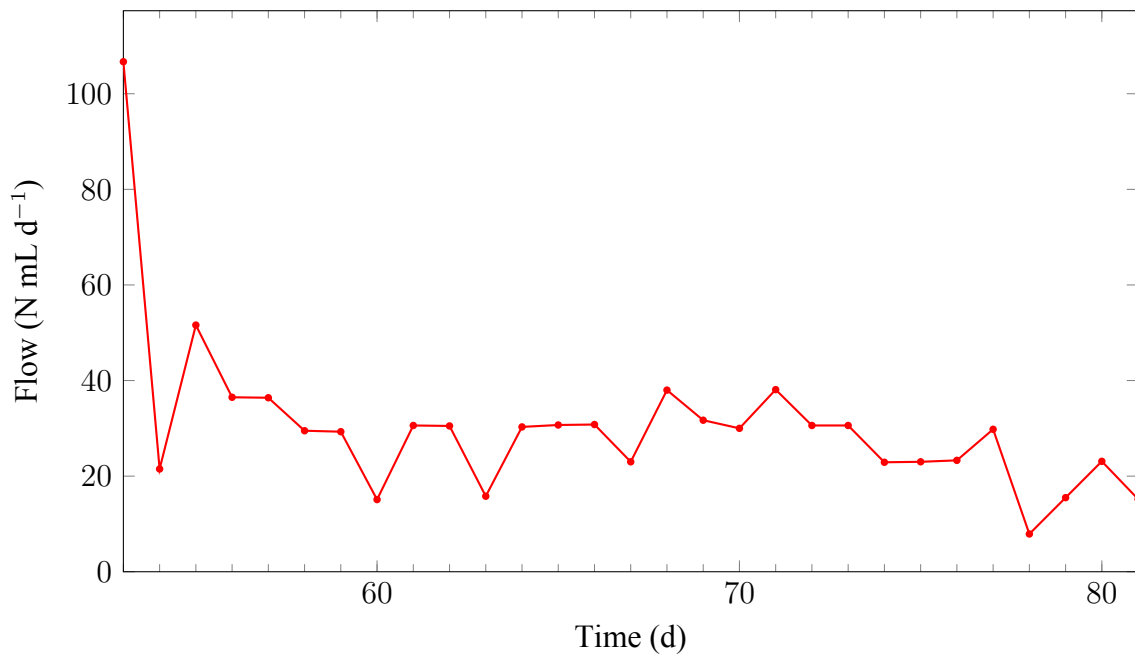


Figure 5.11: Volume flow of CH_4 N mL d^{-1} (Anaerobic digestion of C-GL)

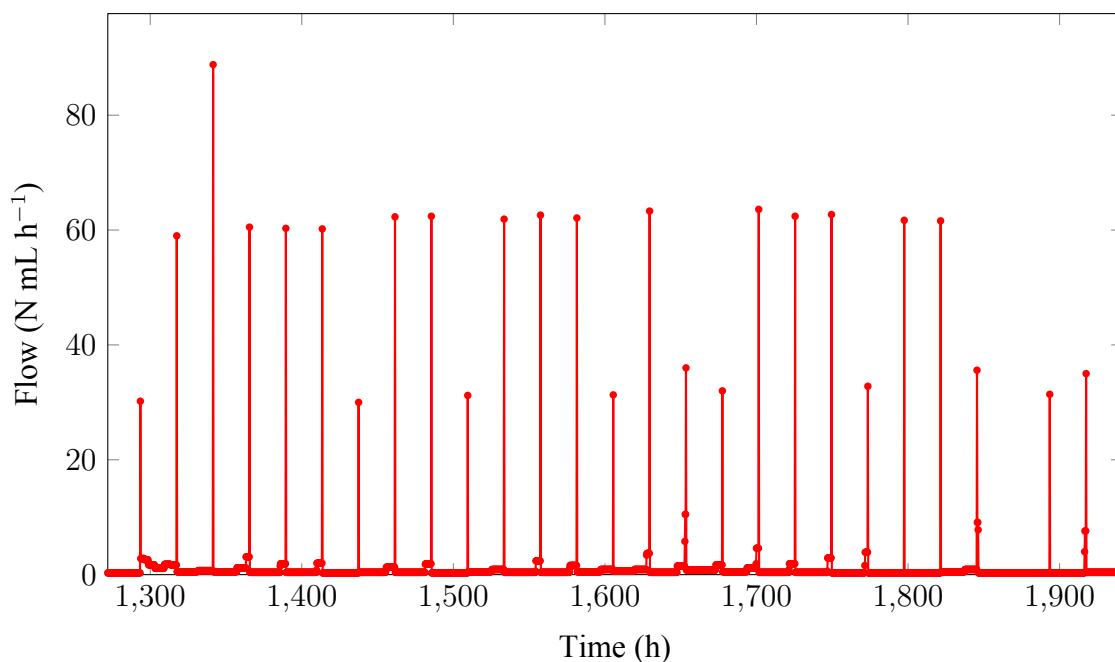


Figure 5.12: Volume flow of CH₄ N mL h⁻¹ (Anaerobic digestion of C-GL)

5.2.5 Post-feeding Period

During the final two weeks, no fresh substrate was added to the reactor, it just had some residual substrate from the C-GL only phase for a week. Nevertheless, a small residual biogas production was recorded (Figs. 5.13 and 5.14). This production can be attributed to the slow degradation of accumulated organic matter and microbial decay products. The flow rate declined gradually, confirming that the reactor was no longer being actively fed and that the remaining substrate was limited.

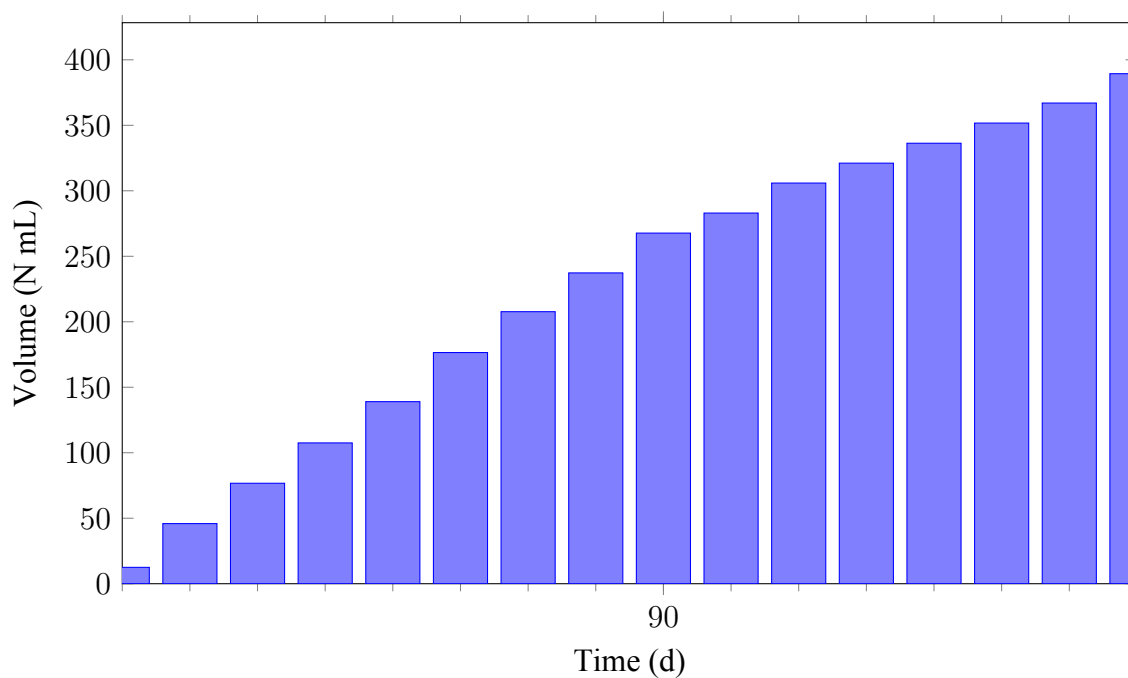


Figure 5.13: Cumulative volume of CH₄ N mL (Post-feeding period)

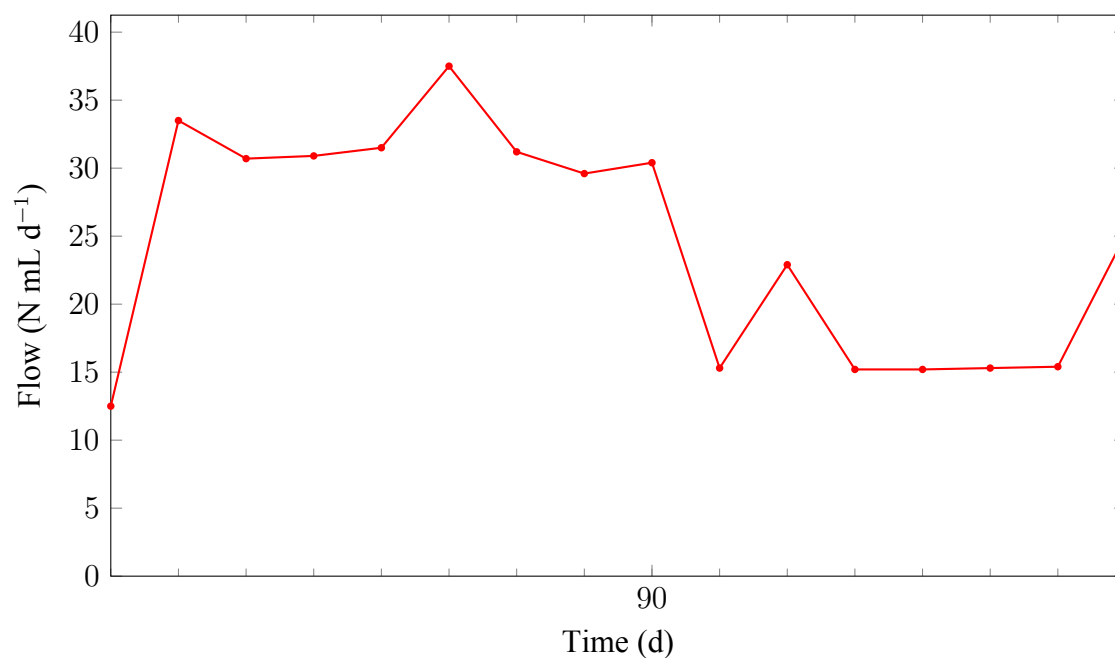


Figure 5.14: Volume flow of CH₄ N mL d⁻¹ (Post-feeding period)

5.2.6 Alkalinity and Volatile Fatty Acids

The total Alk and VFA concentrations were monitored throughout the experiment to assess process stability. The results are shown in Fig. 5.15. Initially, both Alk and VFA were low,

reflecting the early adaptation phase before buffer addition. After the manual introduction of KHCO_3 , Alk increased sharply to values above $7000 \text{ mg CaCO}_3 \text{ L}^{-1}$ and remained relatively high until close to the end. This high Alk was maintained by the addition of bicarbonate to the feed, which helped to counteract the acidity produced during the degradation of the feed.

VFA concentrations also increased after the start of feeding, reaching values between 3000 to $5000 \text{ mg CH}_3\text{COOH L}^{-1}$. Such high VFA levels are often associated with process imbalance, as they can inhibit methanogenesis. However, the high Alk provided a strong buffering capacity, and the pH remained above 6.5 for most of the experiment, with the co-digestion the pH was falling and with C-GL only the pH was increasing. The VFA/Alk is a more reliable indicator of instability. In this experiment, the ratio fluctuated between 0.3 to 1.2 . While using the co-digestion feed a ration around 0.8 to 1.0 had the highest production.

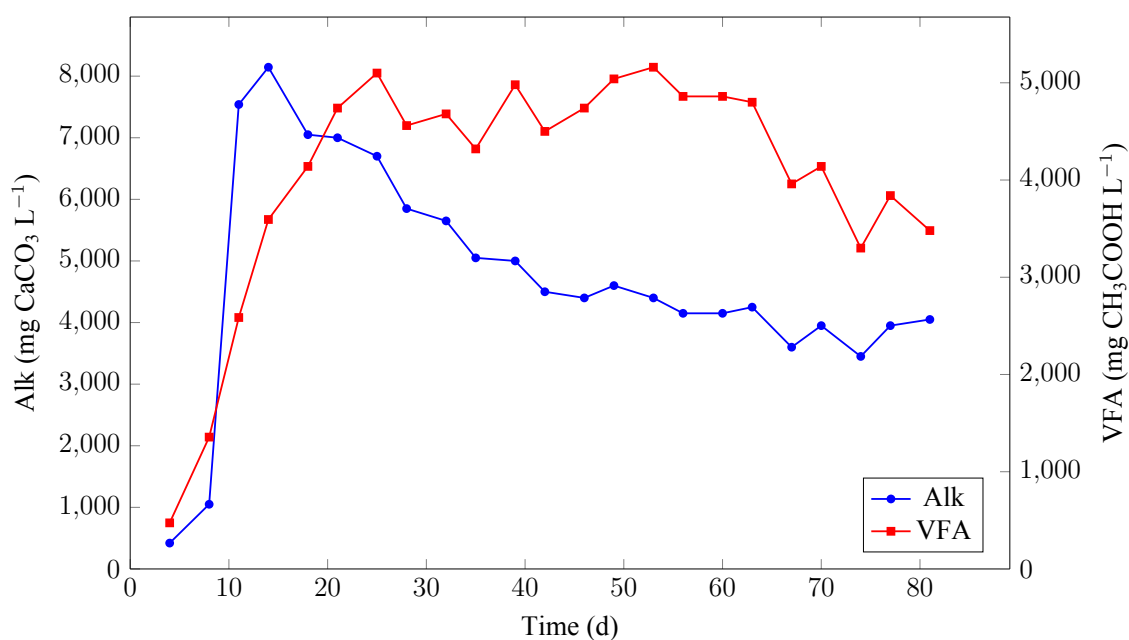


Figure 5.15: Alkalinity & volatile fatty acid comparison graph

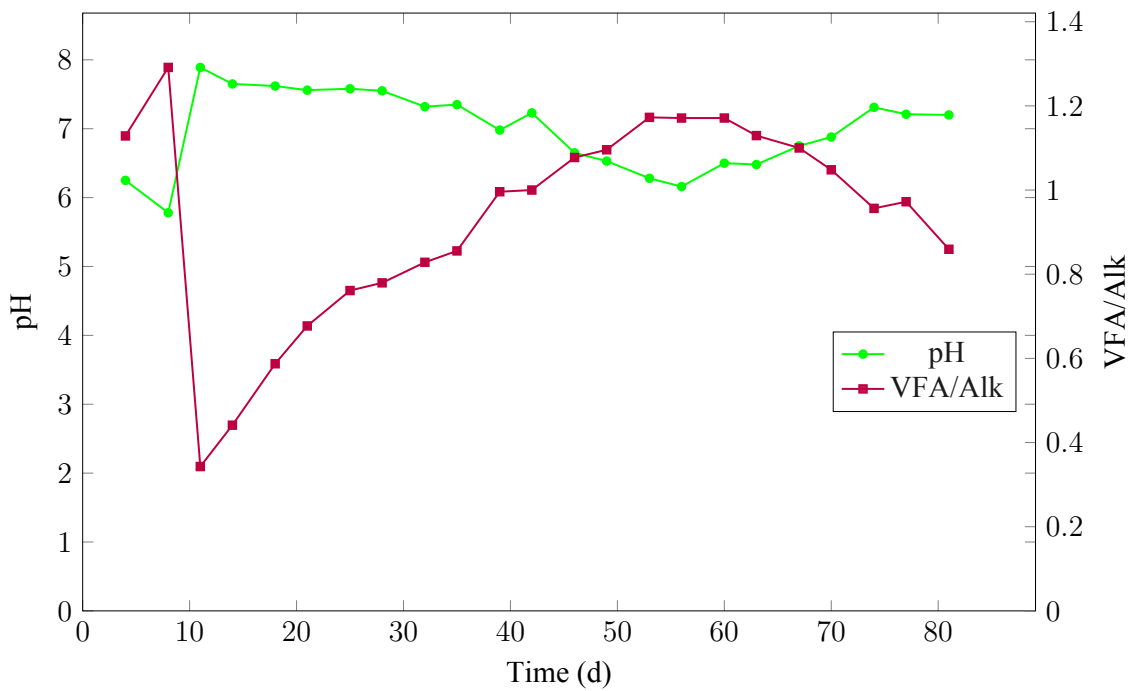


Figure 5.16: pH & VFA/Alk comparison graph

Chapter 6

Conclusion

A direct comparison of the co-digestion and C-GL only phases reveals that the co-digestion of FW and C-GL resulted in higher biogas yields and more stable operation. The pH was decreasing faster with the co-digestion, where with C-GL only phase it wasn't decreasing that fast, and was even growing with the KHCO_3 that was added to maintain the pH. These results support the hypothesis that C-GL, when co-digested with FW, can enhance biogas production by balancing the C/N and providing readily available carbon, whereas C-GL alone may work, but would need a different setup and adjustments.

The C-GL only phase, can't be said to be strictly worse than the co-digestion of FW and C-GL, since both phases were done in sequence in the same reactor, this could have given the C-GL phase disadvantages on a contaminated environment.

Using C-GL, with some P-GL to make the necessary adjustment to balance the C/N, could be a viable solution for the C-GL problem. It's only major problem, is that C-GL contains a lot of impurities that can affect the anaerobic digestion, and for a system that can reliably consume it, it will need a better control of its substrate.

During the whole process, the biggest production of CH_4 was when VFA/Alk was ≈ 1 with a pH of ≈ 6.5 , during the C-GL only phase, even with not optimal values of pH and VFA/Alk, it was able to maintain a reasonable amount of production of CH_4 , showing that it can be a suitable substrate for AD.

Future studies could involve on how to use better the C-GL as a feed, to try subside its downside of its impurities and need for an extra carbon rich substrate to balance the feed.

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